MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, SHRINKABLE WITHOUT HEAT, POLYURETHANE FLEXIBLE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Department and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this specification and the latest isssue of MIL-I-85080.

REQUIREMENTS:

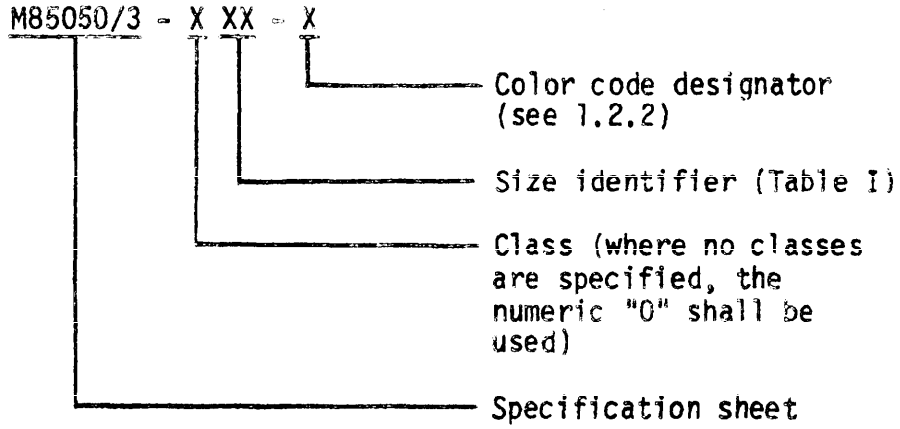
Continuous operating temperature range: -55°C (-67°F) to +125°C (+257°F).

Mode of shrinkage: The heatless shrink tubing shall recover to a predetermined diameter upon exposure to the atmosphere. No external sources of heat are required.

Color: The tubing shall be furnished in black, white or clear as standard colors. Colors shall conform to the requirements of Class II of MIL-STD-104.

Longitudinal change: ± 10 percent.

Military part number: The military part number shall consist of the basic number of this specification sheet and dash numbers as shown below:



Example: White, 0.506 inch as supplied I.D. sleeving shall be identified as M85080/3-007-9.

FSC 5970

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TABLE I. Dimensions (inches). 1/

Military part number 2/	As supplied I.D. min 3/	After unrestricted shrinkage	
		I.D. max	Wall thickness 4/
M85080/3-001-X	0.042	0.031	0.015
M85080/3-002-X	0.084	0.062	0.020
M85080/3-003-X	0.126	0.093	0.020
M85080/3-004-X	0.170	0.125	0.020
M85080/3-005-X	0.252	0.187	0.020
M85080/3-006-X	0.338	0.250	0.025
M85080/3-007-X	0.506	0.375	0.025
M85080/3-008-X	0.675	0.500	0.030
M85080/3-009-X	0.843	0.625	0.030
M85080/3-010-X	1.012	0.750	0.035
M85080/3-011-X	1.350	1.000	0.040

1/ Diameter limits for object to be enclosed should be in accordance with the recommendations of the manufacturer.

2/ The color code identified shall be replaced by the color code from 1.2.2.

3/ The inner diameter of the sleeve may be mechanically expanded 2X the recovered inside diameter when in the as-supplied condition. The manufacturer should be consulted for special applications and recommendations.

4/ Wall shrinkage will be less if shrinkage is restricted.

Unrestricted shrinkage: 4.6.3.12; 23° ± 1°C(73.4° ± 2°F) for 24 hours.

TABLE II. Properties after unrestricted shrinkage.

Property	Requirement	Test condition
Restricted shrinkage	No cracks	4.6.3.2
Voltage withstand	Pass	4.6.6
Dielectric strength, volts per mil min	500	4.6.2; ASTM D 876
Volume resistivity, OHM-CM, min	10^{11}	4.6.2; ASTM D 876
Water absorption, % max	1.0	4.6.2; ASTM D 570, Par 6.1
Fungus resistance	Non nutrient	4.6.2; ASTM G 21
Low temperature flex	No cracking	4.6.6; -55° + 2°C (-67° + 5°F)
Heat shock	No cracks, flowing or dripping	4.6.7; 160° ∓ 1°C (320° ∓ 2°F)
Properties after heat resistance test:		
Tensile strength, psi, min	800	4.6.8; 125° + 1°C (258° ± 2°F) for 96 hours.
Ultimate elongation, %, min	400	
Corrosion	No corrosion or pitting	4.6.9

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TABLE II. Properties after unrestricted shrinkage. - Continued

Property	Requirement	Test condition
Properties after fluid resistance test:		4.6.10
Tensile strength, psi, min	2000	
Ultimate elongation, %, min	300	
Odor	No objectionable odor	4.6.11
Tensile stress, at 200% elongation, psi, max	1000	4.6.2 ASTM D 412
Tensile strength, psi, min	1000	4.6.13 ASTM D 638 Speed D
Ultimate elongation, %, min	300	4.6.13 ASTM D 638 Speed D
Flammability	Self extinguishing; one minute max. No more than 25% of indicator flag burned.	4.6.14.1

Shelf life (as supplied): Test method 4.6.16.

Preparing activity:
Navy - AS

(Project 5970-N590)

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